

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

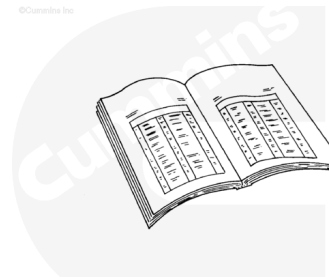
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Drain the coolant. Refer to Procedure 008-018 (Cooling System) in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the aftercooler heat shields. Refer to Procedure 010-129 (Aftercooler Heat Shield) in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Remove the turbocharger heat shields. Refer to Procedure 010-076 (Turbocharger Heat Shield) in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html).
- Remove the turbocharger inlet connection. Refer to Procedure 010-130 (Turbocharger Inlet Connections) in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html)

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- Remove the air cleaner assembly. Refer to Procedure 010-013 (Air Cleaner Assembly (Engine-Mounted)) in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-013-tr.html>)
- Remove the marine gear oil cooler, if equipped. Refer to Procedure 008-041 (Marine Gear Oil Cooler) in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-041-tr.html>)

Remove

with Mechanically Actuated Injector

Remove both turbocharger coolant supply lines (1).

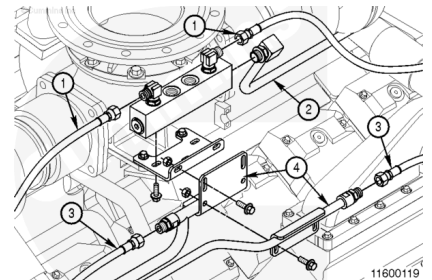
Remove the junction block coolant supply lines (2) and the associated hardware.

Remove the turbocharger oil supply lines (3).

Remove the four capscrews securing the oil supply tube bracket (4).

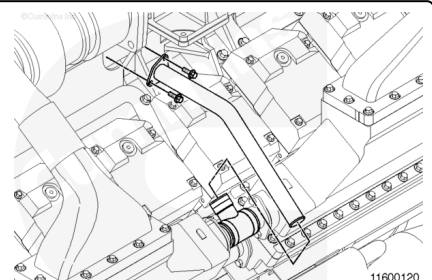
Remove the oil supply tube.

Remove the support brackets securing the coolant junction block and the junction block.



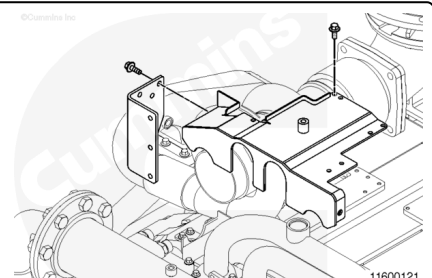
Loosen the clamp securing the rubber tee connections.

Remove the capscrews securing the aftercooler water tube which connects the aftercooler to the exhaust manifold heat shield.



Remove the front exhaust cavity heat shield and front right corner plate capscrews.

Remove the front exhaust cavity heat shield and the front right corner plate.

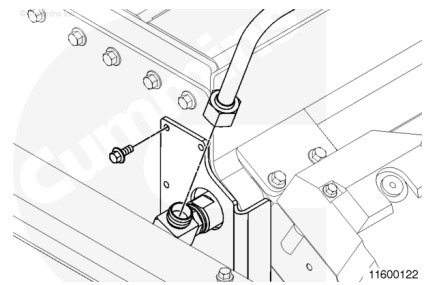


Loosen the water outlet tube nut and remove the tube from the back of the engine.

Note : If the exhaust cavity heat shield does not need to be completely removed from the engine, the fitting does not have to be removed. If the exhaust cavity heat shield has to be completely removed from the engine, the fitting must be removed.

Remove the right rear corner exhaust cavity heat shield capscrews.

Remove the right rear corner exhaust cavity heat shield.



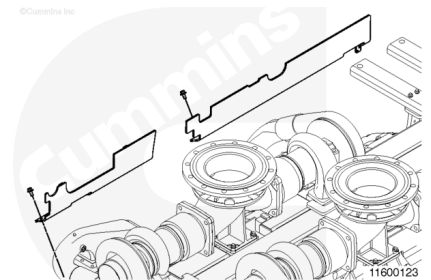
Note : It will possibly be necessary to loosen the exhaust manifold capscrews to allow clearance with the turbocharger heat shield.

Note : It will possibly be necessary to remove left bank rocker covers numbers two, three, six, and seven to allow for removal of the side exhaust cavity heat shields. If the rocker covers are removed, use clean rags to prevent dirt intrusion.

Remove the front and rear side exhaust cavity heat shield capscrews, as well as any necessary coolant piping.

Remove the front and rear side exhaust cavity heat shield.

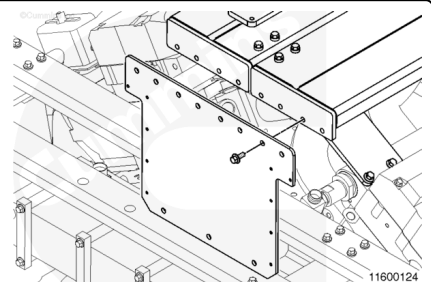
Note : The opposite bank is similar. Use the same steps to remove the opposite bank side exhaust cavity heat shields.



Remove the remaining rear exhaust cavity heat shield capscrews.

Note : Two of the rear exhaust cavity heat shield capscrews are used in the turbocharger oil supply line.

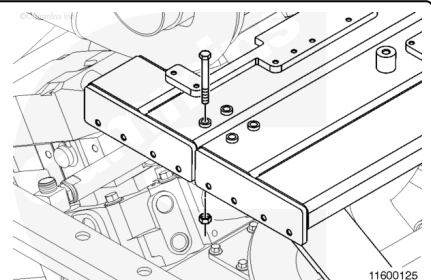
Remove the rear exhaust cavity heat shield.



Remove the left and right bank exhaust manifold heat shield capscrews.

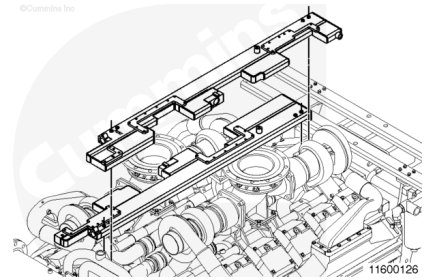
There are four capscrews that have nuts located underneath the left and right bank exhaust manifold heat shield.

Remove the capscrews and nuts in the rear of the left and right bank exhaust manifold heat shields.



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of injury, use a hoist or get assistance to lift this component.



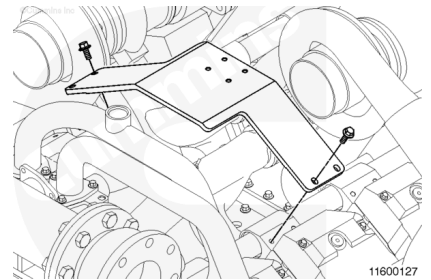
The left bank exhaust manifold heat shield **must** be removed before the right bank exhaust manifold heat shield, because there is a water retention cover that overlaps the right bank exhaust manifold heat shield.

There are two M12 tapped holes that are available for lifting eyes or slings.

Remove the left and right bank exhaust manifold heat shields.

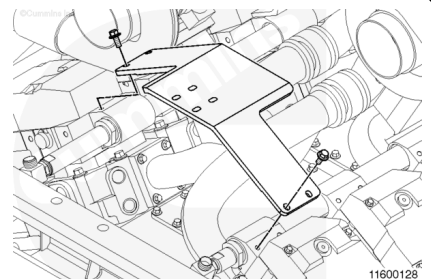
The front exhaust manifold heat shield support bracket has tapped holes and is mounted on the right bank number one and left bank number two rocker lever housings.

Remove the capscrews and the front exhaust manifold heat shield support bracket.



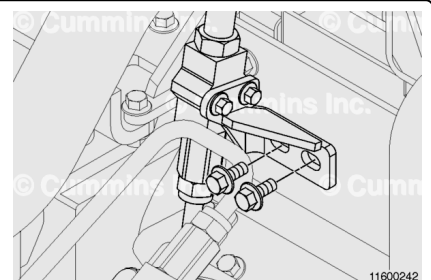
The rear exhaust manifold heat shield support bracket has slotted holes and is mounted on the number eight right and left bank rocker lever housings.

Remove the capscrews and the rear exhaust manifold heat shield support bracket.

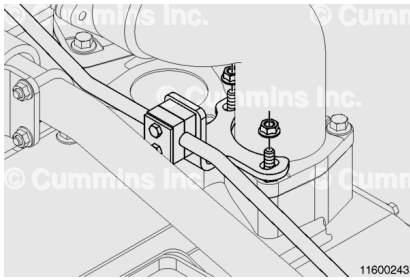


with Electronically Actuated Injector

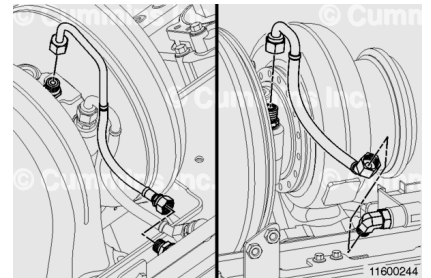
Remove the capscrews securing the high pressure fuel supply line junction block to the left bank exhaust cavity heat shields.



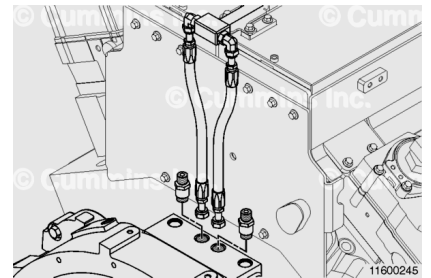
Remove the capscrews securing the right bank high pressure fuel supply line to the mounting bracket, located at the outlet of the low temperature aftercooling (LTA) thermostat housing.



Disconnect the coolant drain fittings and lubricating oil supply from the unions on the turbocharger bearing housing.

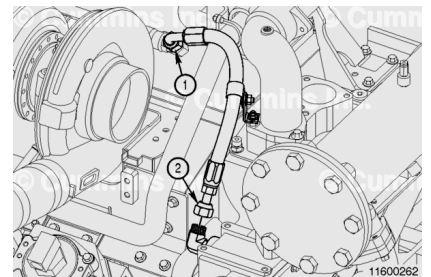


Disconnect the turbocharger lubricating oil supply connections at the flywheel housing.



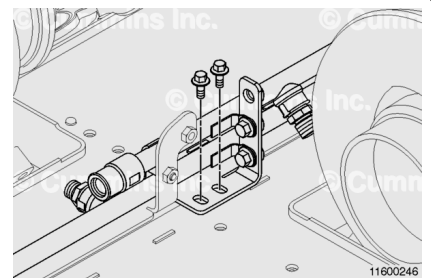
Disconnect the turbocharger coolant drain connection at the right bank thermostat housing.

If applicable, disconnect the marine gear oil cooler coolant supply connection at the low temperature aftercooling (LTA) coolant supply "J" block, located at the left bank front of the engine.



Remove the 16 capscrews securing the turbocharger lubricating oil supply and coolant drain manifold bracket to the exhaust cavity water tank heat shields.

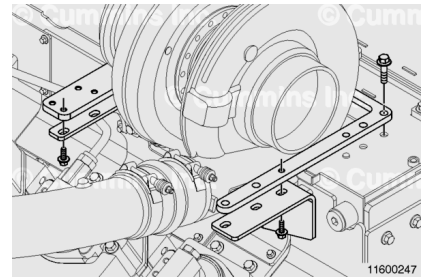
Remove the turbocharger lubricating oil supply and coolant drain manifold and bracket assembly from the engine.



Remove the 8 capscrews securing the turbocharger heat shield support brackets from the left bank front exhaust cavity heat shields and turbocharger heat shield bracket.

Remove the brackets.

Note : The procedure for the right bank is similar. Use the same steps to remove the right bank side turbocharger head shield support brackets.



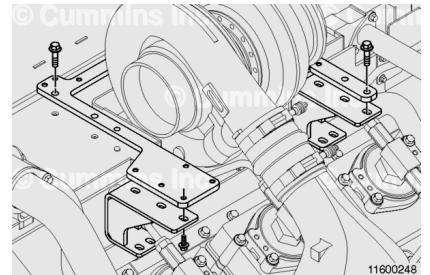
Remove the 14 capscrews securing the turbocharger head shield bracket to the exhaust cavity water tank heat shields at the left bank front turbocharger.

Remove the bracket.

Remove the 13 capscrews securing the turbocharger head shield bracket to the exhaust cavity water tank heat shields at the left bank rear turbocharger.

Remove the brackets.

Note : The procedure for the right bank is similar. Use the same steps to remove the right bank side turbocharger heat shield brackets.

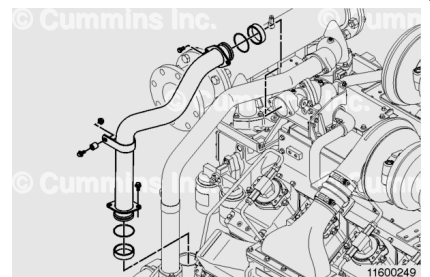


Remove the two capscrews securing the low temperature aftercooling (LTA) supply tube flange to the LTA "J" block casting.

Remove the clip securing the LTA coolant supply tube.

Remove the LTA coolant supply tube from the casting by pivoting the tube away from the "J" block.

Remove the discard the o-ring and wear sleeve.



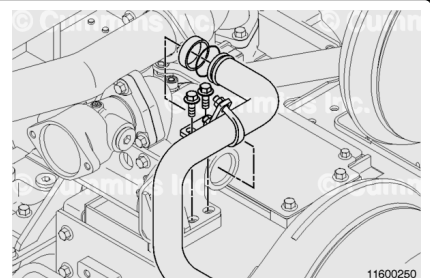
Loosen the clamps securing the left bank low temperature aftercooling (LTA) tubes to the rubber tee connections at the aftercooler.

Remove the two capscrews securing the LTA tube flange to the exhaust cavity water tank heat shield.

Remove the left bank LTA coolant transfer tube from between the aftercooler and the exhaust cavity water tank heat shields.

Remove the u-bolt and nuts securing the left bank LTA supply tube to the "J" block.

Remove the p-clips securing the left bank LTA supply tube to the exhaust cavity side heat shields.



Remove the left bank LTA supply tube by rotating the tube towards the centerline of the engine and pulling the tube from the "J" block.

Remove and discard the o-ring and wear sleeve.

Loosen the clamps securing the right bank LTA tubes to the rubber tee connections at the aftercooler.

Remove the two capscrews securing the LTA tube flange to the exhaust cavity water tank heat shield.

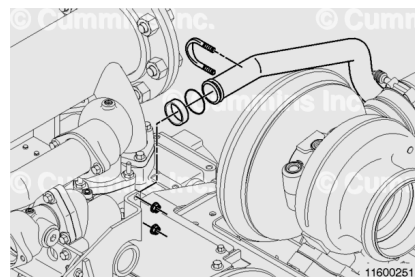
Remove the right bank LTA coolant transfer tube from between the aftercooler and the exhaust cavity water tank heat shields.

Remove the u-bolt and nuts securing the right bank LTA supply tube to the "J" block.

Remove the p-clips securing the right bank LTA supply tube to the exhaust cavity side heat shields.

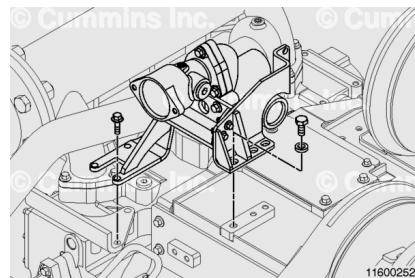
Remove the right bank LTA supply tube by pulling the tube from the "J" block.

Remove and discard the o-ring and wear sleeve.

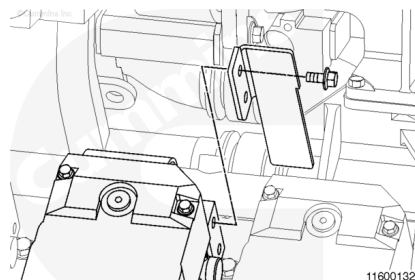


Remove the capscrews securing the "J" block support brackets to the exhaust cavity water tank heat shields and the thermostat housing.

Remove the "J" block and support brackets.



Remove the 2 capscrews securing the small side shields to the rocker housing from between cylinder numbers 4 and 5 on both the right and left banks of the engine.

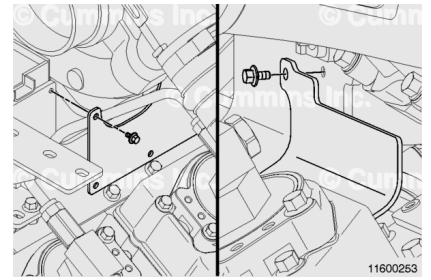


Remove the capscrews securing the left back bank heat shield access plates from the exhaust cavity side shields.

Remove the access plates.

Note : The procedure for the right bank is similar. Use the same steps to remove the right bank side heat

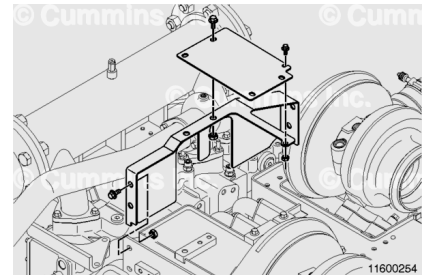
shield access plate.



Remove the 5 capscrews securing the exhaust cavity front heat shield to the left and right bank exhaust cavity side heat shields.

Remove the 5 capscrews securing the exhaust cavity front heat shield and top access plate to the exhaust cavity water tank heat shields.

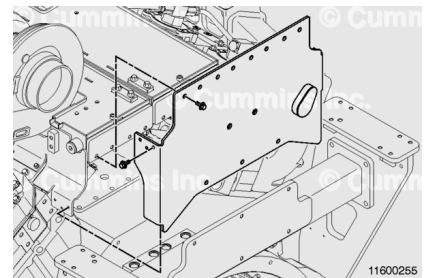
Remove the shields.



Remove the 3 capscrews securing the exhaust cavity rear shield to the left bank exhaust cavity side heat shields.

Remove the 11 capscrews securing the exhaust cavity rear heat shield to the exhaust cavity water tank heat shields and the engine block.

Remove the heat shield.



Remove the 2 capscrews securing the coolant return tube flange to the thermostat housing.

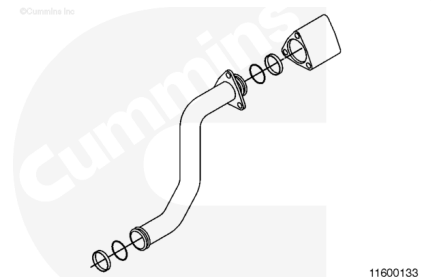
Remove the flange.

Remove the 3 capscrews securing each of the triangular blocks to the exhaust cavity water tank heat shields.

Pull the triangular blocks back onto the coolant return tubes.

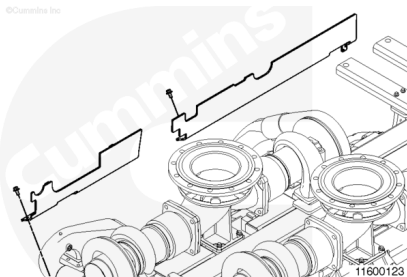
Remove the coolant return tubes from the thermostat housing by pulling them towards the rear of the engine and out of the thermostat housing.

Remove and discard the o-rings from the coolant return tubes.



Remove the capscrews securing the exhaust cavity front and rear side heat shields on both left and right side banks.

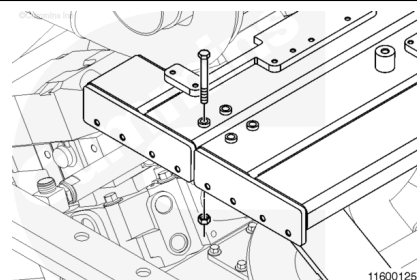
Remove the shields.



Remove the left and right bank exhaust manifold heat shield capscrews.

There are four capscrews that have nuts located underneath the left and right bank exhaust manifold heat shield.

Remove the capscrews and nuts in the rear of the left and right bank exhaust manifold heat shields.



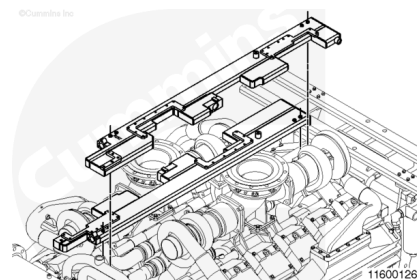
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of injury, use a hoist or get assistance to lift this component.

The left bank exhaust manifold heat shield **must** be removed before the right bank exhaust manifold heat shield, because there is a water retention cover that overlaps the right bank exhaust manifold heat shield.

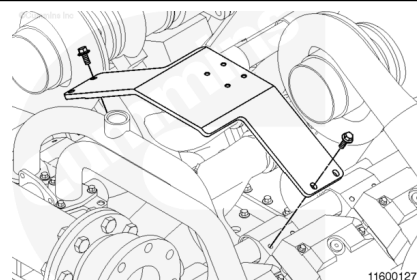
There are two M12 tapped holes that are available for lifting eyes or slings.

Remove the left and right bank exhaust manifold heat shields.



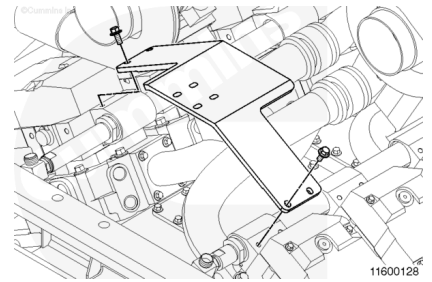
The front exhaust manifold heat shield support bracket has tapped holes and is mounted on the right bank number one and left bank number two rocker lever housings.

Remove the capscrews and the front exhaust manifold heat shield support bracket.



The rear exhaust manifold heat shield support bracket has slotted holes and is mounted on the number eight right and left bank rocker lever housings.

Remove the capscrews and the rear exhaust manifold heat shield support bracket.



Install

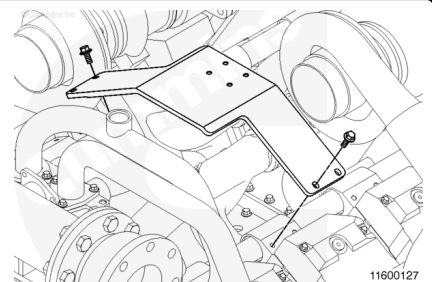
with Mechanically Actuated Injector

When the front exhaust manifold heat shield support bracket is installed onto the engine, the slotted holes **must** face toward the front of the engine.

Place the front exhaust manifold heat shield support bracket onto rocker lever housings right bank one and left bank two.

Install the four M10 x 25 mm capscrews through the bracket and into the rocker lever housing.

Finger tighten the capscrews.

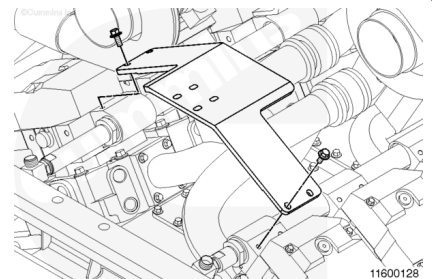


When the rear exhaust manifold heat shield support bracket is installed onto the engine, the slotted holes **must** face toward the rear of the engine.

Place the rear exhaust manifold heat shield support bracket onto the number eight right and left rocker lever housings.

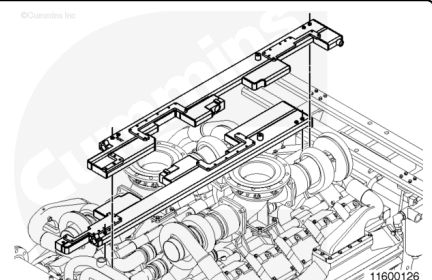
Install the four M10 x 25 mm capscrews through the exhaust manifold heat shield bracket and into the rocker lever housings.

Finger tighten the capscrews.



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of injury, use a hoist or get assistance to lift this component.



The right bank exhaust manifold heat shield **must** be installed first because of the overlapping water retention cover.

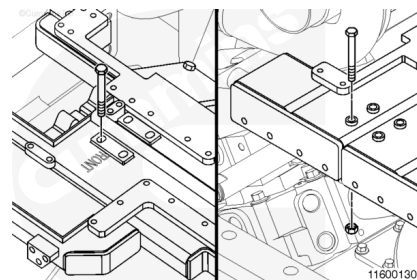
Make sure the water connections are facing towards the front of the engine and the holes in the heat shield line up with those on the right bank side supports, when installing the right and left bank exhaust manifold heat shields.

Place the right bank exhaust manifold heat shield (identified as heat shield with the smaller lip) onto the front and rear heat shield support brackets.

Place the left bank exhaust manifold heat shield onto the front and rear heat shield support brackets.

Install the four M10 x 70 mm capscrews that secure the right and left bank exhaust manifold heat shields to the front support bracket. Finger tighten the capscrews.

Install the four M10 x 80 mm capscrews and nuts that secure the right and left bank exhaust manifold heat shields to the rear support bracket. Finger tighten the capscrews.

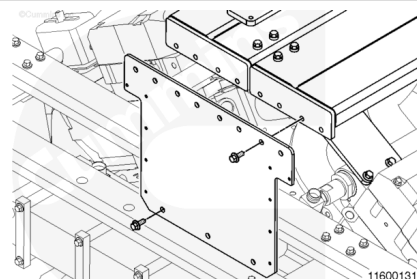


Install the exhaust cavity rear heat shield.

When installing the exhaust cavity rear heat shield capscrews, thread in the capscrews, but do **not** tighten.

Thread the eight M10 x 25 mm upper exhaust cavity rear heat shield capscrews into the exhaust manifold heat shields.

Thread the three M10 x 25 mm lower exhaust cavity rear heat shield capscrews into the engine block.

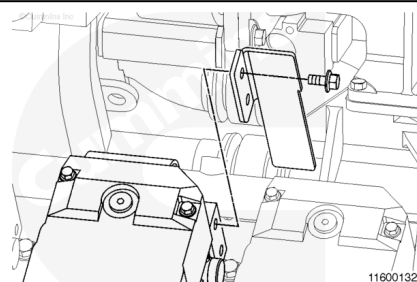


Note : The installation process is similar for the right and left bank rocker lever housing exhaust cavity heat shields.

Install the right and left bank, number four and five rocker lever housing exhaust cavity heat shields.

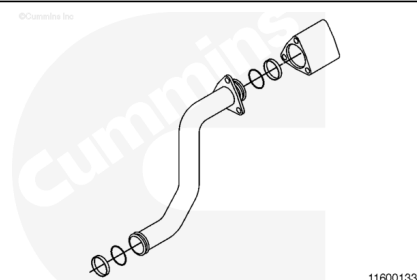
Install and tighten the capscrews.

Torque Value: 45 n·m [33 ft-lb]



Configure the LTA connections for securing to the exhaust manifold heat shields as shown in the graphic. **Only** one S-tube is illustrated in the graphic.

Install the o-ring into the grooves on both ends of the S-tubes.



Fit the wear sleeve over the end of each S-tube, just clear of the o-ring beading.

Apply hellerine to the o-ring on the end with the free moving flange and insert it into the triangular block aligning the flange holes with the block holes.

Install the LTA connections to the exhaust manifold heat shields.

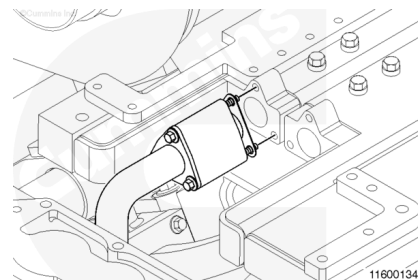
Install the three M8 x 90 mm capscrews through the holes in the flange and the triangular blocks for each connection.

Hang the gasket over the three capscrew ends.

Press the thermostat housing, with the ports facing towards the front of the engine, onto the exhaust manifold heat shield S-tubes.

Tighten the six capscrews.

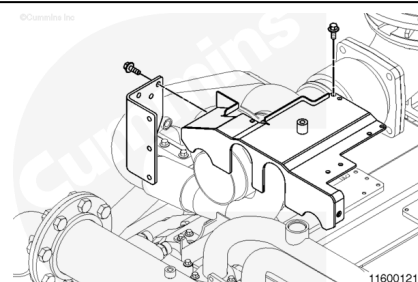
Torque Value: 25 n·m [18 ft·lb]



Install the front heat shield.

Install the nine capscrews that secure the exhaust manifold heat shield to the top exhaust cavity heat shields and the engine block. Finger tighten the capscrews **only**.

Install the right front corner plate and capscrews. Finger tighten the capscrews **only**.



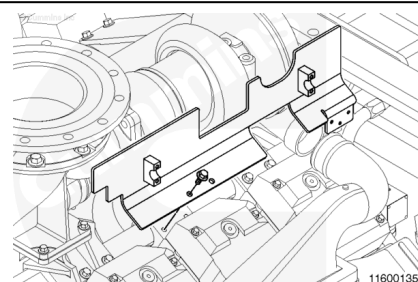
Note : The left bank rear side exhaust cavity heat shield is shown in the graphic. The right bank front side exhaust cavity heat shield is similar.

Install the left bank rear side exhaust cavity heat shield.

Install the two M10 x 25 mm capscrews. Finger tighten the capscrews **only**.

Install the right bank front side exhaust cavity heat shield.

Install the M10 x 25 mm two capscrews. Finger tighten the capscrews **only**.



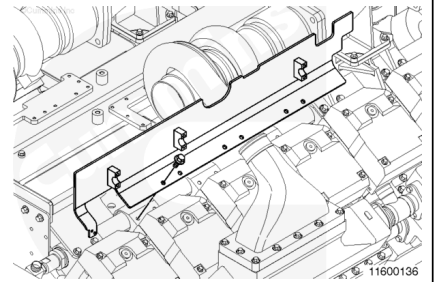
Note : The right bank rear side exhaust cavity heat shield is shown in the graphic. The left bank front side exhaust cavity heat shield is similar.

Install the right bank rear side exhaust cavity heat shield to the rocker lever housings.

Install the six M10 x 25 mm capscrews. Finger tighten the capscrews **only**.

Install the left bank front side exhaust cavity heat shield to the rocker lever housings.

Install the six M10 x 25 mm capscrews. Finger tighten the capscrews **only**.



Install the right bank rear corner plate.

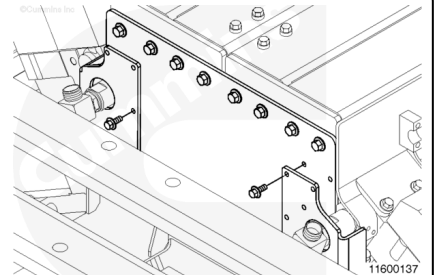
Install the five M6 x 12 mm capscrews.

Install the two M6 x 12 mm capscrews through the right bank rear corner plate into the right bank rear side exhaust cavity heat shield.

Install the left bank rear corner plate.

Install the five M6 x 12 mm capscrews.

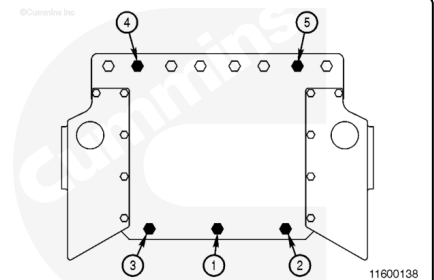
Install the three M6 x 12 mm capscrews through the left bank corner plate into the left bank rear side exhaust cavity heat shield.



Align all brackets with the capscrews.

Tighten capscrews (1) through (5) on the rear heat exhaust cavity heat shield in sequence.

Torque Value: 45 n•m [33 ft-lb]

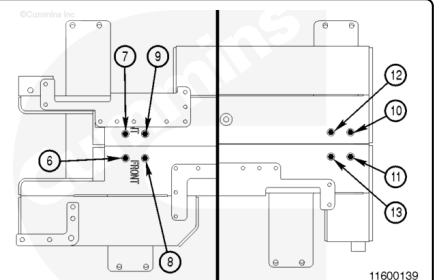


Tighten the four exhaust manifold heat shield front support bracket capscrews, (6) through (9) in sequence.

Torque Value: 45 n•m [33 ft-lb]

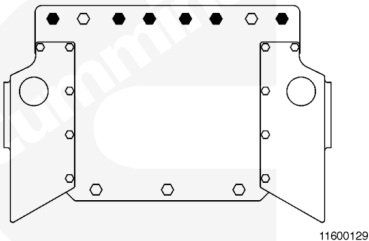
Tighten the four exhaust manifold heat shield rear support bracket capscrews, (10) through (13) in sequence.

Torque Value: 45 n•m [33 ft-lb]



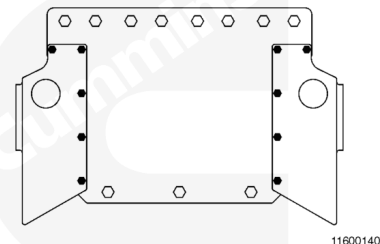
Tighten the remaining rear exhaust manifold heat shield capscrews.

Torque Value: 45 n•m [33 ft-lb]



Tighten the left and right rear corner plate capscrews.

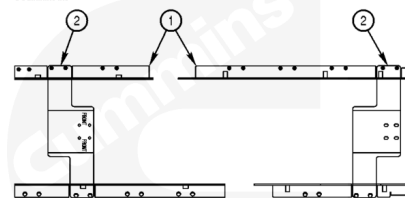
Torque Value: 10 n•m [89 in-lb]



Note : Do not tighten the right front corner plate when completing this step.

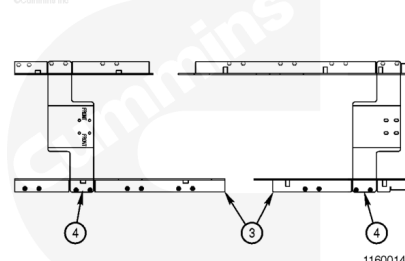
Starting at the rear of the engine, tighten all of the right bank side exhaust cavity heat shield (1) capscrews, and the right bank front and rear exhaust manifold heat shield (2) support bracket capscrews.

Torque Value: 45 n•m [33 ft-lb]



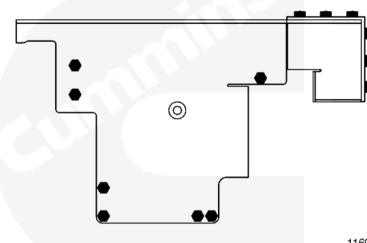
Starting at the front of the engine, tighten all left bank side exhaust cavity heat shield (3) capscrews, and the left bank front and rear exhaust manifold heat shield support bracket (4) capscrews.

Torque Value: 45 n•m [33 ft-lb]



Tighten the front exhaust cavity heat shield capscrews, and front right corner plate capscrews.

Torque Value: 23 n•m [204 in-lb]



Note : Opposite bank is similar, use the same steps to install the opposite bank junction block and associated

hardware.

Install the junction block and mounting brackets.

Tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]

Install the oil supply tube and bracket (4) and associated mounting hardware.

Tighten the capscrews.

Torque Value: 47 n•m [35 ft-lb]

Install and tighten the turbocharger oil supply lines (3).

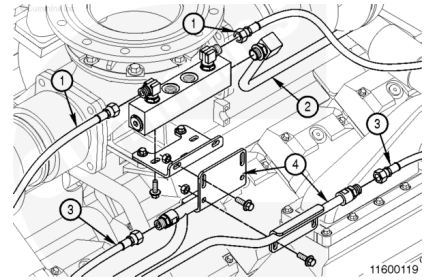
Torque Value: 16 n•m [142 in-lb]

Install and tighten the junction block coolant supply line (2) and the associated mounting hardware.

Torque Value: 16 n•m [142 in-lb]

Install and tighten both turbocharger coolant supply lines (1).

Torque Value: 16 n•m [142 in-lb]

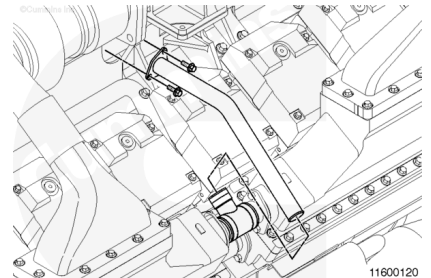


Insert the aftercooler water tube into the rubber tee connection and install the capscrews securing the tube to the exhaust manifold heat shield.

Tighten the capscrews.

Torque Value: 10 n•m [89 in-lb]

Tighten the clamps securing the rubber tee connections.



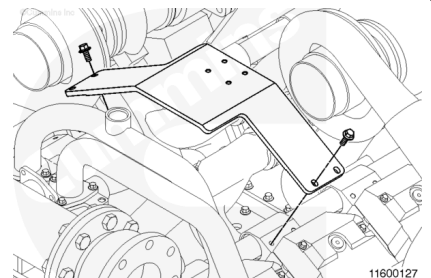
with Electronically Actuated Injector

When the front exhaust manifold heat shield support bracket is installed onto the engine the slotted holes **must** face toward the front of the engine.

Place the front exhaust manifold heat shield support bracket onto rocker lever housings right bank one and left bank two.

Install the four M10 x 25 mm capscrews through the bracket and into the rocker lever housing.

Finger tighten the capscrews.

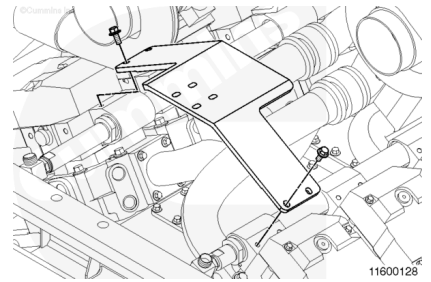


When the rear exhaust manifold heat shield support bracket is installed onto the engine, the slotted holes **must** face toward the rear of the engine.

Place the rear exhaust manifold heat shield support bracket onto the number eight right and left rocker lever housings.

Install the four M10 x 25 mm capscrews through the exhaust manifold heat shield bracket and into the rocker lever housings.

Finger tighten the capscrews.



⚠ WARNING ⚠

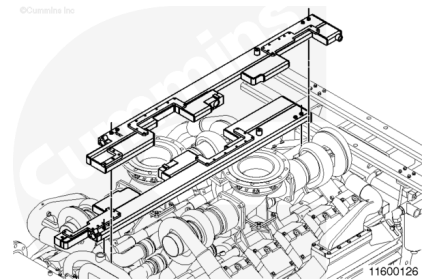
This component weighs 23 kg [50 lb] or more. To reduce the possibility of injury, use a hoist or get assistance to lift this component.

The right bank exhaust manifold heat shield **must** be installed first because of the overlapping water retention cover.

Make sure the water connections are facing towards the front of the engine and the holes in the heat shield line up with those on the right bank side supports, when installing the right and left bank exhaust manifold heat shields.

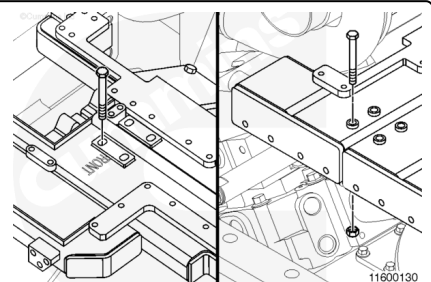
Place the right bank exhaust manifold heat shield (identified as heat shield with the smaller lip) onto the front and rear heat shield support brackets.

Place the left bank exhaust manifold heat shield onto the front and rear heat shield support brackets.



Install the four M10 x 70 mm capscrews that secure the right and left bank exhaust manifold heat shields to the front support bracket. Finger tighten the capscrews.

Install the four M10 x 80 mm capscrews and nuts that secure the right and left bank exhaust manifold heat shields to the rear support bracket. Finger tighten the capscrews.

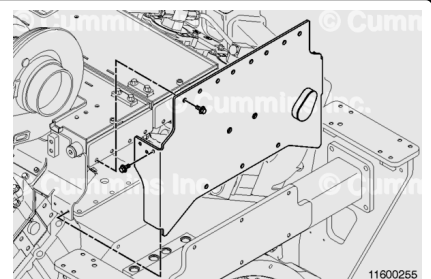


Install the exhaust cavity rear heat shield.

When installing the exhaust cavity rear heat shield capscrews, thread in the capscrews, but do **not** tighten.

Thread the 8 M10 x 25 mm upper exhaust cavity rear heat shield capscrews into the exhaust manifold heat shields.

Thread the 3 M10 x 25 mm lower exhaust cavity rear heat shield capscrews into the engine block.

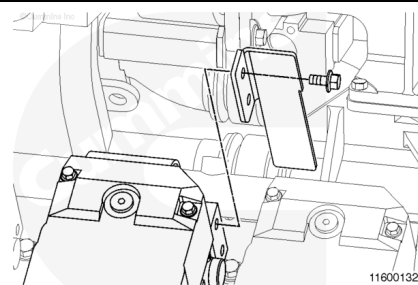


Note : The installation process is similar for the right and left bank rocker lever housing exhaust cavity heat shields.

Install the right and left bank, number four and five rocker lever housing exhaust cavity heat shields.

Install and tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



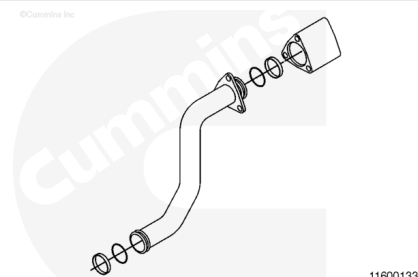
Configure the LTA connections for securing to the exhaust manifold heat shields as shown in the graphic. **Only** one S-tube is illustrated in the graphic.

Install the o-ring into the grooves on both ends of the S-tubes.

Fit the wear sleeve over the end of each S-tube, just clear of the o-ring beading.

Apply hellerrine to the o-ring on the end with the free moving flange and insert it into the triangular block aligning the flange holes with the block holes.

Push the triangular block onto the tube farther than is needed. This will allow for room to install the tube into the thermostat housing.



Install the LTA connections to the exhaust manifold heat shields.

Install the three M8 x 90 mm capscrews through the holes in the flange and the triangular blocks for each connection.

Hang the gasket over the three capscrew ends.

Press the tubes into the thermostat housing until they are firmly seated. Push the triangular blocks up against the exhaust cavity water tank heat shields. The tubes should be centered between the triangular blocks and the thermostat housing.

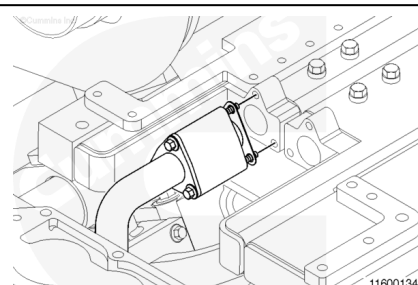
Tighten the six capscrews.

Torque Value: 25 n•m [221 in-lb]

Install the tube flange onto the thermostat housing.

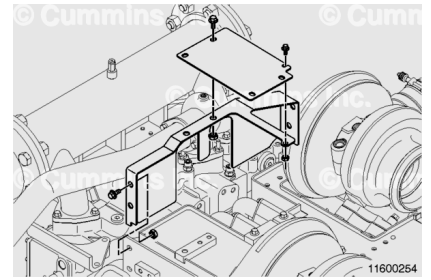
Tighten the 2 capscrews.

Torque Value: 25 n•m [221 in-lb]



Install the exhaust cavity front shield and access cover plate.

Install the 5 capscrews that secure the plates to the exhaust cavity water tank heat shields.



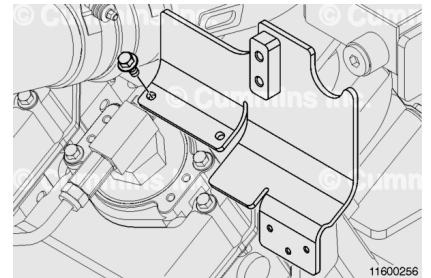
Note : The left bank rear side exhaust cavity heat shield is shown in the illustration. The right bank front side exhaust cavity heat shield is similar.

Install the left bank rear side exhaust cavity heat shield.

Install the two M10 x 25 mm capscrews. Finger tighten the capscrews.

Install the left bank front side exhaust cavity heat shield to the rocker lever housings.

Install the 6 M10 x 25 mm capscrews. Finger tighten the capscrews.



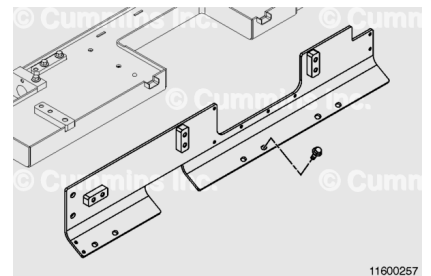
Note : The left bank rear side exhaust cavity heat shield is shown in the illustration. The right bank front side exhaust cavity heat shield is similar.

Install the right bank rear side exhaust cavity heat shield to the rocker lever housings.

Install the 6 M10 x 25 mm capscrews. Finger tighten the capscrews.

Install the left bank front side exhaust cavity heat shield to the rocker lever housings.

Install the 6 M10 x 25 mm capscrews. Finger tighten the capscrews.

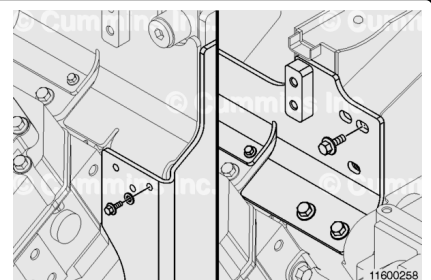


Install the three capscrews that secure the exhaust cavity side heat shields to the exhaust cavity rear heat shield.

Finger-tighten **only**.

Install the 5 capscrews that secure the exhaust cavity front heat shield and access plate to the left and right bank exhaust cavity side heat shields.

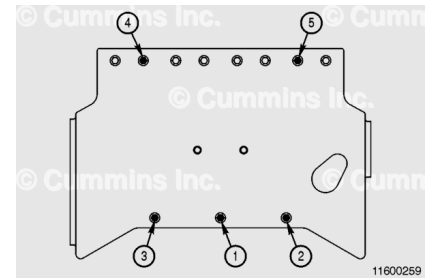
Finger-tighten **only**.



Align all brackets with the capscrews.

Tighten the capscrews (1) through (5) on the rear heat exhaust cavity heat shield in sequence.

Torque Value: 45 n•m [33 ft-lb]

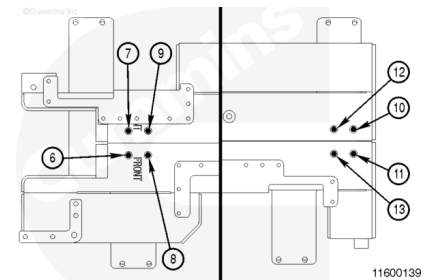


Tighten the 4 exhaust manifold heat shield front support bracket capscrews, (6) through (9) in sequence.

Torque Value: 45 n•m [33 ft-lb]

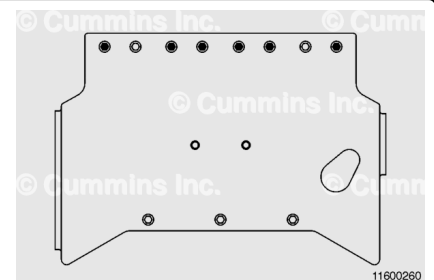
Tighten the four exhaust manifold heat shield rear support bracket capscrews, (10) through (13) in sequence.

Torque Value: 45 n•m [33 ft-lb]



Tighten the remaining rear exhaust manifold heat shield capscrews.

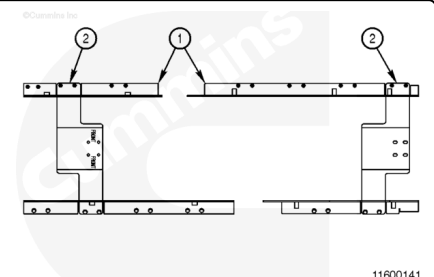
Torque Value: 45 n•m [33 ft-lb]



Note : Do not tighten the right front corner plate when completing this step.

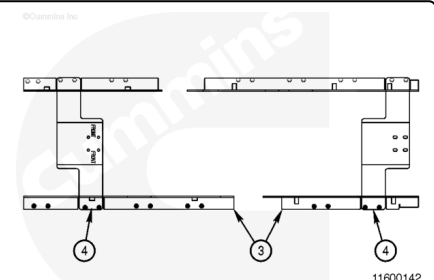
Starting at the rear of the engine, tighten all of the right bank side exhaust cavity heat shield (1) capscrews, and the right bank front and rear exhaust manifold heat shield (2) support bracket capscrews.

Torque Value: 45 n•m [33 ft-lb]



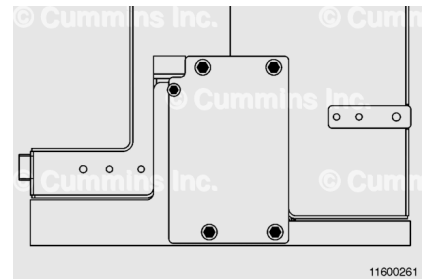
Starting at the front of the engine, tighten all left bank side exhaust cavity heat shield (3) capscrews, and the left bank front and rear exhaust manifold heat shield support bracket (4) capscrews.

Torque Value: 45 n•m [33 ft-lb]



Tighten the front exhaust cavity heat shield capscrews.

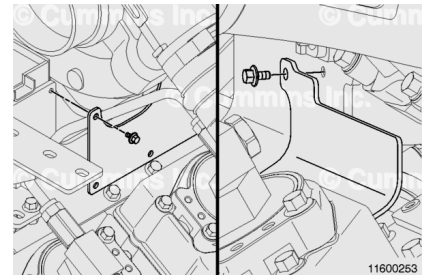
Torque Value: 23 n•m [204 in-lb]



Install the left bank side heat shield access plates.

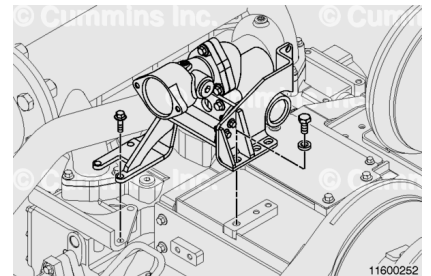
Install the capscrews securing the left bank side heat shield access plates from the exhaust cavity side shields.

Note : The procedure for the right bank is similar. Use the same steps to remove the right bank side heat shield access plates.



Install the LTA "J" block and support brackets.

Install the capscrews securing the "J" block support brackets to the exhaust cavity water tank heat shields and the thermostat housing.



Install new o-rings and wear sleeves onto the LTA tubes.

Install the right bank LTA supply tube by pressing the tube into the "J" block.

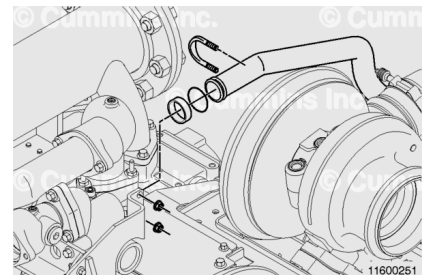
Install the p-clips that secure the LTA supply tube to the turbocharger heat shield brackets.

Install the u-bolt and nuts securing the right bank LTA tube on the "J" block.

Install the LTA transfer tube between the aftercooler and the exhaust cavity water tank heat shields.

Install the 2 capscrews that secure the LTA coolant transfer tube in the exhaust cavity water tank heat shield.

Tighten the clamps securing the right bank LTA tubes to the rubber tee connections at the aftercooler.



Install new o-rings and wear sleeves onto the LTA tubes.

Install the left bank LTA supply tube by pressing the tube into the “J” block and rotating the tube to meet the rubber tee connection at the aftercooler.

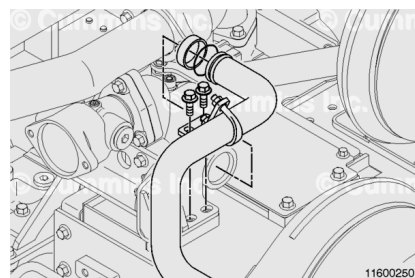
Install the p-clips that secure the LTA supply tube to the turbocharger heat shield brackets.

Install the u-bolt and nuts securing the left bank LTA tube to the “J” block.

Install the LTA transfer tube between the aftercooler and the exhaust cavity water tank heat shields.

Install the capscrews that secure the LTA coolant transfer tube in the exhaust cavity water tank heat shield.

Tighten the clamps securing the left bank LTA tubes to the rubber tee connections at the aftercooler.

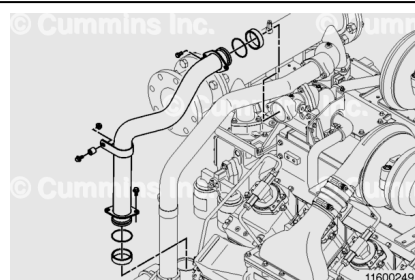


Install new o-rings and wear sleeves onto the LTA supply tube.

Install the LTA coolant supply tube by pressing the tube into the casting and then rotating and pivoting the tube into the “J” block casting.

Install the clips that secure the LTA coolant supply tube.

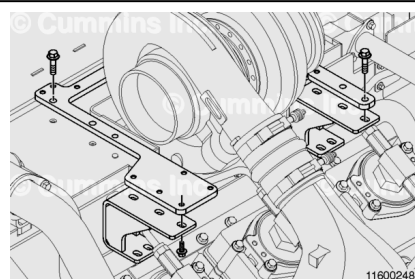
Install the 2 capscrews securing the LTA supply tube flange to the LTA “J” block casting.



Install the turbocharger heat shield bracket and the 14 capscrews securing the turbocharger heat shield bracket to the exhaust cavity water tank heat shields at the left bank front turbocharger.

Install the turbocharger heat shield bracket and the 13 capscrews securing the turbocharger heat shield bracket to the exhaust cavity water tank heat shields at the left bank rear turbocharger.

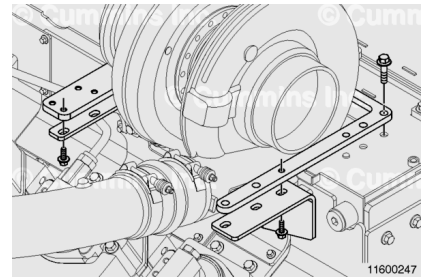
Note : The procedure for the right bank is similar. Use the same steps to install the right bank side turbocharger heat shield brackets.



Install the 8 capscrews securing the turbocharger heat shield support brackets from the left front bank exhaust cavity heat shields and turbocharger heat shield bracket.

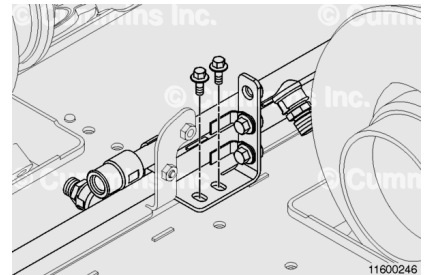
Install the 8 capscrews securing the turbocharger heat shield support brackets from the left bank rear exhaust cavity heat shields and turbocharger heat shield bracket.

Note : The procedure for the right bank is similar. Use the same steps to install the right bank side turbocharger heat shield brackets.



Install the turbocharger lubricating oil supply and coolant drain manifold and bracket assembly onto the engine.

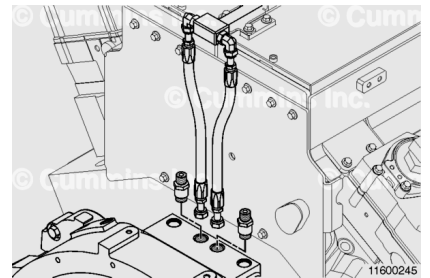
Install the 16 capscrews securing the turbocharger lubricating oil supply and coolant drain manifold bracket to the exhaust cavity water tank heat shields.



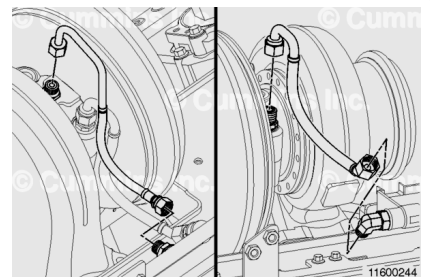
Connect the turbocharger lubricating oil supply connections at the flywheel housing.

Connect the turbocharger coolant drain connection at the right bank thermostat housing.

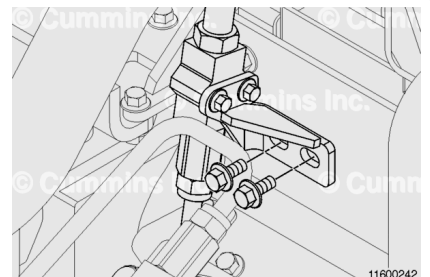
If applicable, connect the marine gear oil cooler coolant supply connection at the LTA coolant supply "J" block, located at the left bank front of the engine.



Connect the lubricating oil supply and coolant drain fittings to the unions on the turbocharger bearing housing.



Install the capscrews securing the high pressure fuel supply line junction block to the left bank exhaust cavity heat shields.



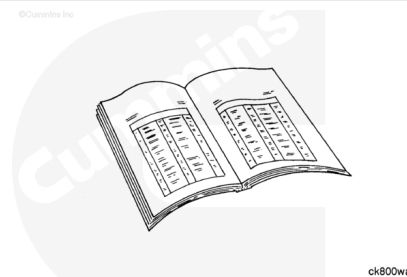
Install the capscrews securing the right bank high pressure fuel supply line to the mounting bracket located at the outlet of the LTA thermostat housing.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the marine gear oil cooler, if equipped. Refer to Procedure 008-041 (Marine Gear Oil Cooler) in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-041-tr.html>)
- Install the air cleaner assembly. Refer to Procedure 010-013 (Air Cleaner Assembly (Engine-Mounted)) in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-013-tr.html>)
- Install the turbocharger inlet connections. Refer to Procedure 010-130 (Turbocharger Inlet Connections) in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html>)
- Install the turbocharger heat shields. Refer to Procedure 010-076 (Turbocharger Heat Shield) in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>)
- Install the aftercooler heat shields. Refer to Procedure 010-129 (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>).
- Fill the engine with coolant. Refer to Procedure 008-018 (Cooling System) in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Operate the engine and check for leaks.

